Name: ____________________________ Date: ________________ Period: _________

Keep in Mind

This is a practice test. It mirrors the real Unit 5 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Rational function **D**
2. Domain restriction **A**
3. Vertical asymptote **H**
4. Hole **E**
5. Horizontal asymptote **B**
6. Least common denominator **G**
7. Extraneous solution **F**
8. Constant of variation **C**

(A) A value that makes the *original* denominator zero, so it can never be an input.

(B) A horizontal line the graph approaches at its far left and far right; found by comparing the degrees of the numerator and denominator.

(C) The number k in $y = kx$ or $y = \frac{k}{x}$; you find it from one known pair of values.

(D) A quotient of two polynomials whose denominator is not zero.

(E) A single missing point on an otherwise unbroken curve, left behind by a factor that cancels.

(F) A number that turns up when you solve but must be thrown out, because it is an excluded value of the original equation.

(G) The smallest expression that every denominator divides evenly, built from the factored denominators.

(H) A vertical line the graph runs alongside but never touches, left by a factor that does *not* cancel.

Part B — Multiple Choice (12 pts)

Circle the best answer.

1. The domain of $f(x) = \dfrac{x+5}{x^2-9}$ is all real numbers except (A) $x = -5$ (B) $x = 3$ **(C)** $x = \pm 3$ $\leftarrow$ (D) none — all reals

2. Simplified completely, $\dfrac{x^2-25}{x^2+3x-10}$ is (A) $\dfrac{x-5}{x-2}$, $x \neq 2$ **(B)** $\dfrac{x-5}{x-2}$**,** $x \neq -5, 2$ $\leftarrow$ (C) $\dfrac{x+5}{x-2}$, $x \neq 2$ (D) $\dfrac{-5}{3x-10}$

3. The horizontal asymptote of $f(x) = \dfrac{3x^2-1}{x^2+4x}$ is (A) $y = 0$ (B) $y = \frac{1}{3}$ **(C)** $y = 3$ $\leftarrow$ (D) there is none

4. Which statement about $h(x) = \dfrac{x^2-x-6}{x^2-9}$ is true?
   (A) There are vertical asymptotes at $x = 3$ and at $x = -3$.
   (B) There is a hole at $x = 3$ and a vertical asymptote at $x = -3$. $\leftarrow$
   (C) There is a hole at $x = -3$ and a vertical asymptote at $x = 3$.
   (D) There is a hole at $x = -2$ and a vertical asymptote at $x = 3$.

5. The graph of $g(x) = \dfrac{1}{x-4} + 2$ has asymptotes **(A)** $x = 4$**,** $y = 2 \leftarrow$ (B) $x = -4$, $y = 2$ (C) $x = 4$, $y = 0$ (D) $x = 2$, $y = 4$

6. y varies *inversely* as x, and $y = 8$ when $x = 3$. When $x = 12$, $y =$ **(A)** $2 \leftarrow$ (B) 6 (C) 24 (D) 32

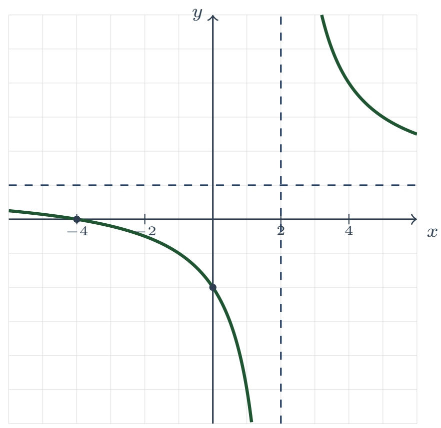
Teacher Note

Item 1: set the *denominator* to zero — $x^2 - 9 = 0$ gives $x = \pm 3$; (A) is the numerator-zero error and (B) keeps only one of the two. Item 2: $\frac{(x-5)(x+5)}{(x+5)(x-2)}$. (A) is the signature Unit 5 error — restrictions read off the *simplified* denominator, losing the hole at $x = -5$; (D) cancels the x^2 terms. Item 3: equal degrees, so the horizontal asymptote is the ratio of leading coefficients, $\frac{3}{1}$; (A) is the $n < m$ rule, (B) inverts the ratio. Item 4: $\frac{(x-3)(x+2)}{(x-3)(x+3)}$ — the $(x-3)$ cancels (hole), the $(x+3)$ does not (wall). (A) skips the cancel test entirely; (D) reads a restriction off the numerator. Item 5: $x - 4$ shifts *right*, so the wall moves to $x = 4$; $+2$ lifts the horizontal asymptote to $y = 2$. Item 6: $k = xy = 24$, so $y = \frac{24}{12} = 2$. (D) is the direct-variation answer, (C) reports k itself.

Part C — Short Answer & Computation (40 pts)

Show your work. State every restriction on the variable.

1. **Read the graph.** The rational function f is graphed below; the dashed lines are its asymptotes and each gridline is one unit.



(a) Equation of the vertical asymptote: $x = 2$ Equation of the horizontal asymptote: $y = 1$

(b) x-intercept: $(-4, 0)$; y-intercept: $(0, -2)$

(c) Domain (interval notation): $(-\infty, 2) \cup (2, \infty)$

(d) Range (interval notation): $(-\infty, 1) \cup (1, \infty)$

(e) End behavior: as $x \to \pm\infty$, $f(x) \to 1$

(f) Interval(s) where f is decreasing: $(-\infty, 2)$ **and** $(2, \infty)$ **— decreasing on *each* branch, but not on their union**

(g) Interval(s) where $f(x) > 0$: $(-\infty, -4) \cup (2, \infty)$

2. **Simplify and state the restrictions.** Read the restrictions from the *original* denominator.

(a) $\dfrac{3x^2 - 27}{x^2 + x - 12}$ (b) $\dfrac{x^2 - 16}{4 - x}$

(a) GCF first, then difference of squares: $\dfrac{3(x-3)(x+3)}{(x+4)(x-3)} = \dfrac{3(x+3)}{x+4}$**,** $x \neq 3,\ x \neq -4$**. The restriction** $x \neq 3$ **survives even though the factor cancelled — that is the hole.**

(b) $4 - x = -(x-4)$**, so** $\dfrac{(x-4)(x+4)}{-(x-4)} = -(x+4)$**,** $x \neq 4$**. Opposite binomials leave a factor of** -1 **behind.**

3. **Multiply or divide.** Factor first; give the simplest form with its restrictions.

(a) $\dfrac{x^2-4}{x^2+6x+9}\cdot\dfrac{x+3}{x-2} = \dfrac{(x-2)(x+2)}{(x+3)^2}\cdot\dfrac{x+3}{x-2} = \dfrac{x+2}{x+3},\ \ x \neq -3,\ x \neq 2.$ **Cancelling runs across the whole product — the $(x-2)$ in the first numerator meets the $(x-2)$ in the second denominator.**

(b) $\dfrac{x^2-x-6}{x^2-1} \div \dfrac{x-3}{x+1}$ **Keep–change–flip:** $\dfrac{(x-3)(x+2)}{(x-1)(x+1)}\cdot\dfrac{x+1}{x-3} = \dfrac{x+2}{x-1},\ \ x \neq 1,\ x \neq -1,\ x \neq 3.$ **Three sources of restriction: the original denominator (± 1), the divisor's denominator (-1 again), and the divisor's *numerator* ($x = 3$, because it becomes a denominator after the flip).**

4. **Add, subtract, and simplify a complex fraction.**

(a) $\dfrac{5}{x+2} - \dfrac{3}{x-1}$ **LCD $= (x+2)(x-1)$:** $\dfrac{5(x-1)-3(x+2)}{(x+2)(x-1)} = \dfrac{5x-5-3x-6}{(x+2)(x-1)} = \dfrac{2x-11}{(x+2)(x-1)},\ \ x \neq -2,\ x \neq 1.$ **The subtraction hits *both* terms of $3(x+2)$.**

(b) $\dfrac{x}{x^2-9} + \dfrac{2}{x+3}$ **Factor first: $x^2-9 = (x-3)(x+3)$, so the LCD is already $(x-3)(x+3)$:** $\dfrac{x+2(x-3)}{(x-3)(x+3)} = \dfrac{3x-6}{(x-3)(x+3)} = \dfrac{3(x-2)}{(x-3)(x+3)},\ \ x \neq \pm 3.$ **Nothing cancels — $(x-2)$ is not a factor of either denominator factor.**

(c) $\dfrac{\frac{1}{x}+\frac{1}{2}}{\frac{1}{x}-\frac{1}{2}}$ **Combine top and bottom over the common denominator $2x$, then divide:** $\dfrac{\frac{2+x}{2x}}{\frac{2-x}{2x}} = \dfrac{2+x}{2x}\cdot\dfrac{2x}{2-x} = \dfrac{2+x}{2-x},\ \ x \neq 0,\ x \neq 2.$ **The $x \neq 0$ comes from the little fractions and would be invisible in the final answer.**

5. **The parent function and its transformations.** Let $g(x) = \dfrac{1}{x+3} - 2$.

(a) Describe the transformation(s) that take $y = \dfrac{1}{x}$ to g. **left 3 units, down 2 units**

(b) Vertical asymptote: $x = -3$; horizontal asymptote: $y = -2$

(c) Domain: $x \neq -3$; range: $y \neq -2$

(d) A different hyperbola of the form $y = \dfrac{1}{x-h} + k$ has vertical asymptote $x = 1$ and horizontal asymptote $y = 4$. Write its equation. $y = \dfrac{1}{x-1} + 4$

6. **Analyze from the equation.** For $f(x) = \dfrac{2x^2-8}{x^2-x-6}$:

(a) Factor the numerator and denominator, and list every excluded value. $f(x) = \dfrac{2(x-2)(x+2)}{(x-3)(x+2)}$**. The original denominator is zero at $x = 3$ and $x = -2$, so both are excluded.**

(b) Which excluded value gives a *hole* and which gives a *vertical asymptote*? Give the coordinates of the hole. **$(x+2)$ cancels, so $x = -2$ is a hole; $(x-3)$ stays, so $x = 3$ is a vertical asymptote. Height of the hole from the simplified form $\dfrac{2(x-2)}{x-3}$ at $x = -2$: $\dfrac{2(-4)}{-5} = \dfrac{8}{5}$, so the hole is $\left(-2, \frac{8}{5}\right)$.**

(c) Horizontal asymptote: $y = 2$ (Justify with a degree comparison.) **Numerator and denominator are both degree 2, so the asymptote is the ratio of leading coefficients, $\frac{2}{1} = 2$.**

(d) x-intercept: $(2, 0)$; y-intercept: $\left(0, \frac{4}{3}\right)$

7. **Solve. Check every answer against the excluded values.**

(a) $\dfrac{x}{x+2} + \dfrac{1}{x-1} = 1$ **Excluded: $x \neq -2, 1$. Multiply by the LCD $(x+2)(x-1)$: $x(x-1)+(x+2) = (x+2)(x-1)$, so $x^2+2 = x^2+x-2$, giving $x = 4$. Since 4 is not excluded, $x = 4$ is the solution.**

(b) $\dfrac{x}{x-4} = \dfrac{4}{x-4} + 3$ **Excluded: $x \neq 4$. Multiply by $(x-4)$: $x = 4+3(x-4) = 3x-8$, so $-2x = -8$ and $x = 4$. But 4 is exactly the excluded value — it is extraneous. The equation has no solution.**

(c) $\dfrac{x}{x-2} + \dfrac{2}{x} = \dfrac{4}{x^2-2x}$ **Factor the last denominator: $x^2-2x = x(x-2)$, so the LCD is $x(x-2)$ and**

the excluded values are $x \neq 0, 2$. Multiplying through: $x^2 + 2(x-2) = 4$, so $x^2 + 2x - 8 = 0$ and $(x+4)(x-2) = 0$, giving $x = -4$ or $x = 2$. Throw out $x = 2$ (excluded). Only $x = -4$ survives; check: $\frac{-4}{-6} + \frac{2}{-4} = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$ and $\frac{4}{24} = \frac{1}{6}$ ✓

8. **Variation.** The table below records a pair of related quantities.

x	2	4	5	10
y	30	15	12	6

(a) Is y directly or inversely proportional to x? Give the computation from the table that proves it.
Inversely — the *products* are constant: $2 \cdot 30 = 4 \cdot 15 = 5 \cdot 12 = 10 \cdot 6 = 60$ (the ratios y/x are not)

(b) Find the constant of variation k and write the equation. **$k = 60$, so $y = \dfrac{60}{x}$**

(c) Use your equation to find y when $x = 8$. $y = \dfrac{60}{8} = 7.5$

Teacher Note

Item 1(f) is the Unit 5 habit-breaker: the answer is two *separate* intervals, never $(-\infty, \infty)$ minus a point written as one interval — nothing crosses a wall. Item 1(d): accept "all reals except $y = 1$." Items 2–4: a simplified form with *no* restriction list is half an answer — score it as such; the restriction must come from the *original* denominator. Item 3(b) is the three-sources item: watch for the missing $x \neq 3$ from the divisor's numerator. Item 6(b): the hole's height must come from the *simplified* form; substituting into the original gives $\frac{0}{0}$. Item 7(b) is the payoff loop — "$x = 4$" as a final answer earns no credit; the check against the excluded values is the graded step. Accept "no solution" or "∅". Item 8(a): a student who computes y/x and finds it changing has still shown the right work, provided they then check the products.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The full analysis.** Consider $f(x) = \dfrac{x^2 - 5x + 4}{x^2 - 16}$. **(a)** Factor both parts and list every excluded value of the *original* function. **(b)** Run the cancel test: which excluded value is a hole and which is a vertical asymptote? Give the hole's coordinates and explain how you found its height. **(c)** State the horizontal asymptote, justify it with a degree comparison, and write the end behavior it describes. **(d)** Give the x- and y-intercepts. **(e)** Write the domain in interval notation. **(f)** A classmate simplifies f to $\dfrac{x-1}{x+4}$ and says the two functions are the same. Say exactly where they agree and where they do not. **(a) $f(x) = \dfrac{(x-1)(x-4)}{(x-4)(x+4)}$. The original denominator is zero at $x = 4$ and $x = -4$, so both are excluded.**

(b) $(x-4)$ cancels, so $x = 4$ is a *hole*; $(x+4)$ does not, so $x = -4$ is a *vertical asymptote*. The hole's height comes from the simplified form $\dfrac{x-1}{x+4}$ evaluated at $x = 4$: $\dfrac{3}{8}$. The hole is $(4, \frac{3}{8})$. Substituting 4 into the *original* would give $\frac{0}{0}$, which is why the simplified form is the one to use.

(c) Both degrees are 2, so the horizontal asymptote is the ratio of leading coefficients, $\frac{1}{1} = 1$: the line $y = 1$. That is the end-behavior statement — as $x \to -\infty$ and as $x \to +\infty$, $f(x) \to 1$.

(d) x-intercept: the simplified numerator is zero at $x = 1$, giving $(1, 0)$. y-intercept: $f(0) = \dfrac{4}{-16} = -\dfrac{1}{4}$, giving $(0, -\frac{1}{4})$.

(e) $(-\infty, -4) \cup (-4, 4) \cup (4, \infty)$.

(f) They agree at every input both accept — every x other than 4 and -4. They do not agree at $x = 4$: the simplified expression happily returns $\frac{3}{8}$ there, but $f(4)$ does not exist. Two equivalent expressions can have different *domains*, so cancelling never restores an input.

2. **Modeling.** A club pays a one-time setup fee of \$180 to a screen printer, plus \$7 for each shirt printed. The

average cost per shirt for an order of n shirts is

$$A(n) = \frac{7n + 180}{n}.$$

(a) Explain why A is a rational function, and state the domain that makes sense for this situation. **(b)** Find $A(20)$ and $A(60)$, with units. What is happening to the average cost? **(c)** Rewrite $A(n)$ in the form $7 + \frac{180}{n}$ and use it to state the horizontal asymptote. What does that number mean about this order? **(d)** Find the order size for which the average cost is exactly \$9 per shirt. **(e)** Can the average cost ever be exactly \$7? Answer using the horizontal asymptote. **(a) It is a quotient of two polynomials, $7n + 180$ over n. Algebraically the only restriction is $n \neq 0$; in context n must be a positive whole number of shirts, so the sensible domain is $n = 1, 2, 3, \ldots$ — an order of 0 shirts has no "per shirt" cost at all.**

(b) $A(20) = \frac{140 + 180}{20} = \frac{320}{20} = \$16$ per shirt; $A(60) = \frac{420 + 180}{60} = \frac{600}{60} = \$10$ per shirt. The average cost is *falling* as the order grows, because the one-time \$180 is being split among more shirts.

(c) Divide term by term: $A(n) = \frac{7n}{n} + \frac{180}{n} = 7 + \frac{180}{n}$. As $n \to \infty$ the second piece shrinks toward 0, so the horizontal asymptote is $y = 7$. That \$7 is the true printing cost of one shirt — the setup fee per shirt is the whole difference between the average cost and \$7.

(d) Solve $\frac{7n + 180}{n} = 9$. Multiply by n (allowed, since $n \neq 0$): $7n + 180 = 9n$, so $180 = 2n$ and $n = 90$ shirts.

(e) No. $A(n) = 7 + \frac{180}{n}$, and $\frac{180}{n}$ is positive for every positive n, so $A(n) > 7$ always. The graph approaches the line $y = 7$ but never reaches it — the setup fee never fully disappears, it only gets spread thinner.

Teacher Note

Scoring Part D (6 pts each). D1: 1 pt (a) both factorings with *both* excluded values from the original denominator; 2 pts (b) hole at $x = 4$ *and* wall at $x = -4$, with the height $\frac{3}{8}$ taken from the simplified form (1 pt only if the two are named but the height is missing or computed from the original); 1 pt (c) $y = 1$ with the degree comparison, not just the answer; 1 pt (d) both intercepts; 1 pt (e) the three-piece union — a domain written as one interval, or missing the $x \neq 4$, loses it. (f) is the concept check: fold it into (b)/(e) credit if you prefer, but a student who says the two functions are simply "the same" has missed the unit's central idea. D2: 1 pt (a) rational *and* a context domain of positive whole numbers; 1 pt (b) both values with dollar units and the observation that cost is falling; 2 pts (c) the rewrite *and* $y = 7$ interpreted as the real per-shirt printing cost (1 pt for $y = 7$ alone); 1 pt (d) $n = 90$; 1 pt (e) "no," justified by $\frac{180}{n} > 0$ or by the asymptote never being reached. Do not accept "because it's an asymptote" with no reason attached.